Table 2. Spatially validated predictive performance of MCRK and benchmark regressors.

RF (X + B) is the reference; all other models use X + B + S.

Model	2016		2022		2026	
	R^2	RMSE	R^2	RMSE	R^2	RMSE
RF (X + B)	0.345	2.957	**0.294**	**1.222**	0.319	0.860
RF	0.312 (−9.6%)	3.031 (−2.5%)	0.279 (−5.3%)	1.236 (−1.1%)	0.361 (+13.0%)	0.835 (+2.9%)
Cubist	0.289 (−16.2%)	3.081 (−4.2%)	0.127 (−57.0%)	1.358 (−11.1%)	0.273 (−14.5%)	0.891 (−3.6%)
XGBoost	0.246 (−28.8%)	3.170 (−7.2%)	0.263 (−10.6%)	1.249 (−2.2%)	0.358 (+12.1%)	0.837 (+2.7%)
RBF−SVM	**0.423 (+22.5%)**	**2.776 (+6.1%)**	0.105 (−64.3%)	1.375 (−12.5%)	0.355 (+11.3%)	0.837 (+2.7%)
GBM	0.270 (−21.8%)	3.122 (−5.6%)	0.275 (−6.5%)	1.239 (−1.4%)	0.310 (−2.9%)	0.868 (−0.9%)
KNN	0.328 (−5.1%)	2.999 (−1.4%)	0.141 (−52.1%)	1.348 (−10.3%)	0.077 (−75.8%)	1.000 (−16.3%)
Elastic net	0.268 (−22.4%)	3.127 (−5.7%)	0.125 (−57.3%)	1.360 (−11.3%)	0.025 (−92.1%)	1.020 (−18.6%)
MCRK	0.419 (+21.5%)	2.786 (+5.8%)	0.287 (−2.6%)	1.229 (−0.5%)	**0.628 (+96.8%)**	**0.635 (+26.1%)**

Parentheses: relative R2 change and RMSE reduction versus RF (X + B); negative values indicate deterioration.

Scores are means of five repeat−level spatial cross−validation metrics. Percentages use unrounded scores.

Bold: highest R^2 or lowest RMSE in each campaign, including the reference. RMSE: μg m^{-3}.